Queueing Dynamics Shape Power Demand in AI Data Centers

Subir Majumder[1,2]

[1] Harvard John A. Paulson School of Engineering and Applied Sciences (SEAS)
[2] Department of Electrical and Computer Engineering, Texas A&M University

Contact:
subir-em@ieee.org

Why this matters

Most grid studies still rely on annual energy use, average utilization, or stylized hourly load shapes.

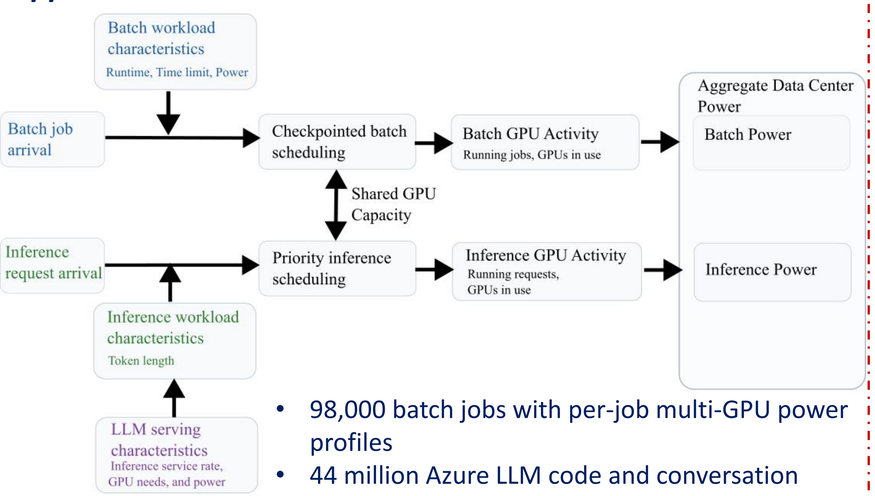
Those approaches miss the high-frequency demand dynamics that matter for grid operations, interconnection, and flexibility planning.

AI data centers are not exogenous loads. Their power demand emerges from workload arrivals, queueing, scheduling, and GPU activation.

Research question

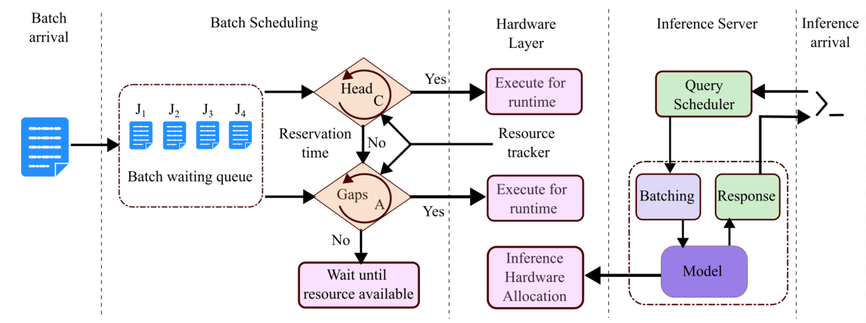
How do batch training and online inference workloads translate into power variability and ramping when they share a common GPU pool?

Approach

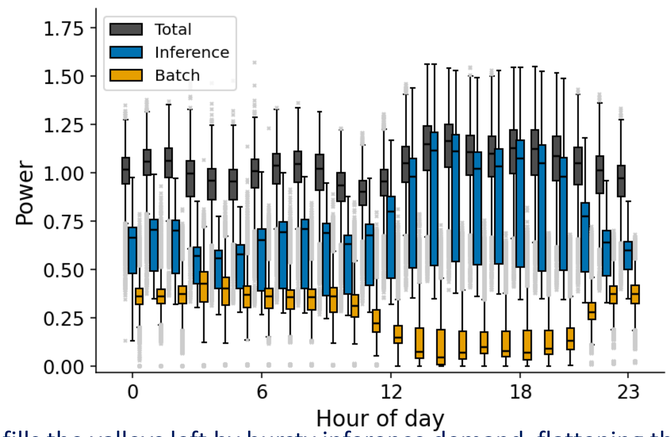


- 98,000 batch jobs with per-job multi-GPU power profiles
- 44 million Azure LLM code and conversation requests
- independent serving measurements mapping token throughput to GPU demand and power

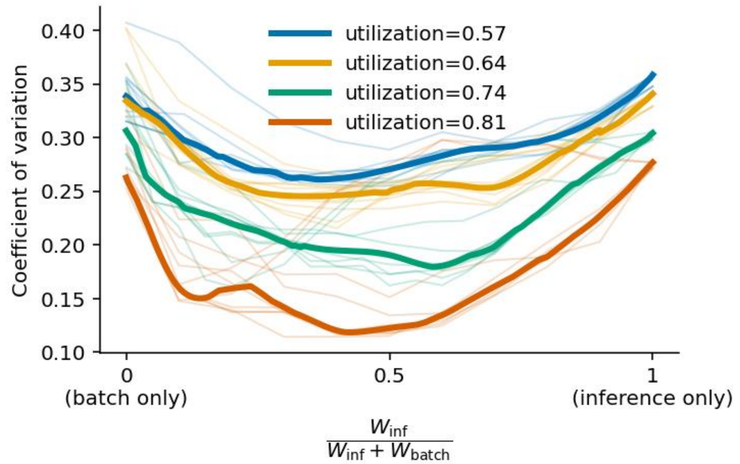
System schematic



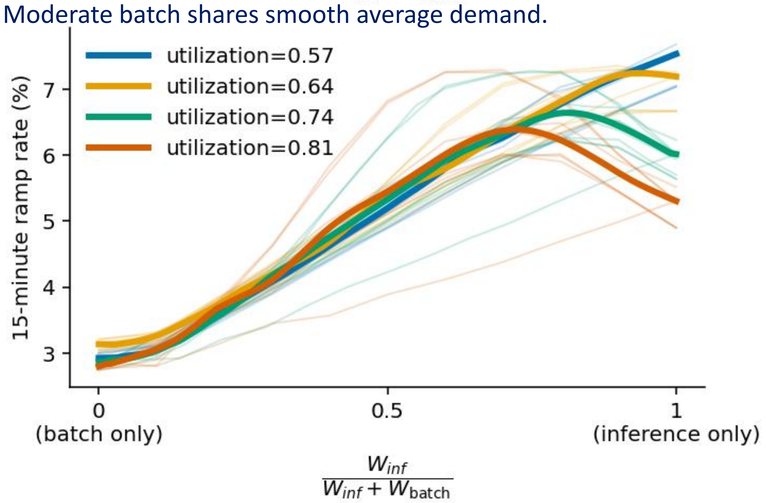
Workload composition changes power nonlinearly



Batch fills the valleys left by bursty inference demand, flattening the daily power profile



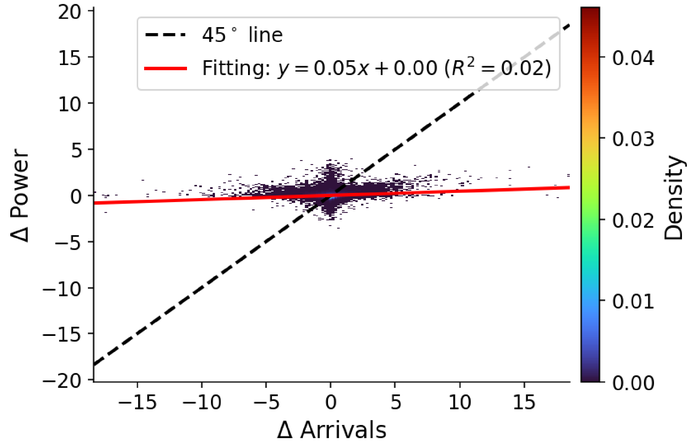
Moderate batch shares smooth average demand.



Intermediate workload mixes can produce stronger 15-minute ramps.

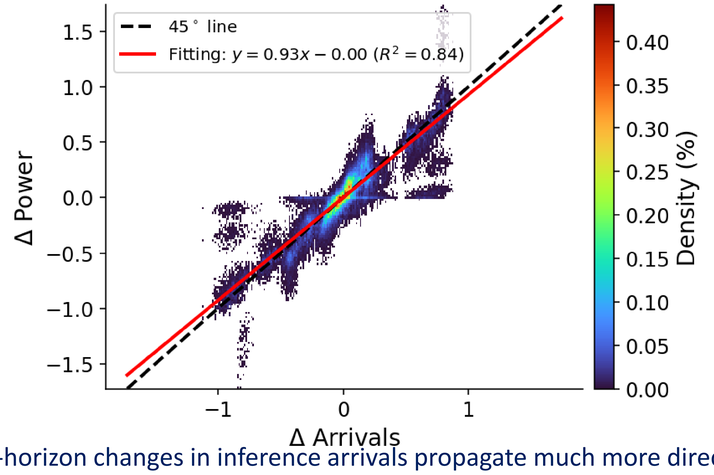
A data center can therefore appear smoother overall while becoming harder for the grid to serve.

Why batch smooth?



Short-horizon variation in batch arrivals transmits only weakly into realized power. Jobs can wait in queue, so bursty demand is absorbed before it reaches execution.

Why inference transmits bursts more directly?



Short-horizon changes in inference arrivals propagate much more directly into realized power. The main buffering margin is request residence time: slower service or longer outputs smooth power by keeping requests active for longer.

Key message for planners and operators

- AI data centers are a new class of electric load whose grid-facing behavior emerges from internal service dynamics.
- Planning models that rely on average utilization or stylized hourly flexibility constraints can miss both the timing and operational difficulty of serving AI demand.

References:

1. Subir Majumder, Minlan Yu and Le Xie. "Workload Composition and Queueing Shape AIData Center Power Demand."